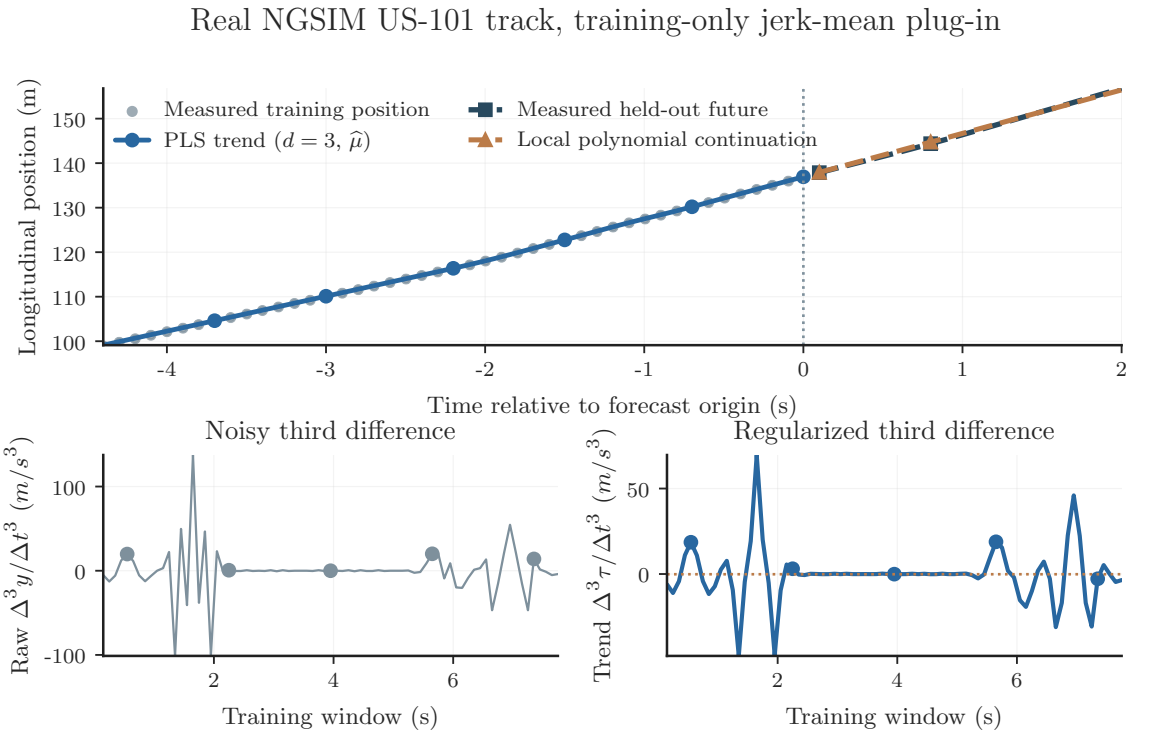

Real NGSIM US-101 track, training-only jerk-mean plug-in
Measured training position
PLS trend ($d = 3$, $\widehat{\mu}$)
Measured held-out future
Local polynomial continuation
Longitudinal position (m)
150
140
130
120
110
100
-4
-3
-2
-1
0
1
2
Time relative to forecast origin (s)
Noisy third difference
Raw $\Delta^3 y/\Delta t^3$ (m/s^3)
100
0
-100
2
4
6
Training window (s)
Regularized third difference
Trend $\Delta^3 \tau/\Delta t^3$ (m/s^3)
50
0
2
4
6
Training window (s)